

When Better Choice Models Do Not Matter: An Empirical Audit of Large-Scale Pairwise Preference Ranking

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Abstract

Leaderboards such as Chatbot Arena rank language models with Bradley–Terry, a logistic-link model whose independence-of-irrelevant-alternatives assumption is widely criticized. It seems natural that a richer Thurstonian link—flexible noise, native ties—should do better. We test this belief with a pre-registered audit on 1.67 million Arena votes, replicated on a second 2025 release, and find it false. We also show why it could not have been otherwise. Before fitting anything, we compute an *effect ceiling*: on the ability gaps that real matchups occupy, no plausible link in the family can beat the logistic by more than a third of one practically meaningful unit. The data respect that ceiling everywhere. Refit stability under sixteen months of pool growth is indistinguishable. Held-out calibration is practically equivalent. Tie mechanisms with provably different shapes fit ties equally well. What separates the models instead is structure, not fit: a single-width tie band buys tie mass with decisive accuracy, while Davidson’s tie parameter is exactly orthogonal to its link (Theorem 1). A steeper link’s apparent cold-start protection dissolved under audit into a numerical artifact; explicit ridge regularization delivers the real thing. One anomaly resists attribution: temporally localized excess dispersion under every link tested. The broader lesson is simple. Richer links help only when approximation error dominates estimation error—and in large, drifting, cold-start-ridden preference systems, it does not.

1 Introduction

Preference leaderboards face a standing accusation. Chatbot Arena [3] ranks language models by maximum-likelihood Bradley–Terry (BT) [2], and BT’s logistic link commits it to Luce’s independence of irrelevant alternatives [12]. Human preferences owe that axiom nothing. The comparison pool grows every month, ties are a fifth of quality votes and outside BT’s outcome space, and every refit re-estimates a global model over a graph that has changed shape. The natural remedy is also classical: Thurstonian random-utility models [17], made computationally exact for arbitrary noise densities by Cotton’s lattice construction [4], with ties arising natively as dead-heat mass rather than by parametric patch [5].

So the hypothesis writes itself. *A richer, tie-aware Thurstonian link should produce better rankings than BT.* To our knowledge nobody had tested it at scale. We test it on 1.67 million Arena votes spanning 129 models and sixteen months, under pre-registered designs: effect thresholds derived from deployment-meaningful errors, synthetic detection gates passed before real data, verdict criteria fixed in advance. We then replicate every comparison on an independent 2025 Arena release.

The hypothesis is false. Refit stability, held-out calibration, and tie fit are practically indistinguishable between the two families, in both datasets. More importantly, we can say *why*, and could have said it before fitting: a computation we call an effect ceiling shows that on the empirical distribution of matchup gaps, the best possible advantage of any link in this Thurstonian family over the logistic is $0.23\text{--}0.31\times$ our minimum practical difference (Section 4). The experiments did not

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merely fail to find an effect. There was provably almost no effect to find. The thesis of this paper, which every section reinforces, is:

Richer preference links improve rankings only when approximation error dominates estimation error — and in a large, evolving preference system, it does not.

What dominates instead is measurable. All methods share the same large next-month miscalibration from nonstationary drift. Up to three quarters of next-month votes involve a model no ability estimator can score. Where the families do differ, the difference is structural rather than statistical: Davidson’s tie parameter provably never touches its decisive link, while the lattice’s single width parameter buys tie mass with decisive accuracy (Theorem 1, measured in Section 6.4). And the one apparent counterexample we observed—a steep link protecting against a cold-start blow-up—turned out, under our own audit machinery, to be a numerical artifact; explicit ridge regularization provides the genuine version of that protection (Section 6.3.1).

Our contributions are three. **(1) A pre-registered, replicated negative result:** link choice has no practical effect on stability, calibration, or tie fit in a real large-scale preference deployment. **(2) A reusable methodology:** effect ceilings computed before fitting, synthetic gates before data, and pattern-based verdict criteria fixed in advance—the discipline that made the negative result trustworthy and that caught our own artifact. **(3) A structural account of why, and of what matters instead:** a tie-orthogonality theorem and its entanglement converse, regularization rather than link shape as the cold-start lever, the dominance of drift and coverage—plus one honest, unattributed anomaly (Section 7).

2 Related work

Paired comparisons. Thurstone’s random-utility model [13, 17] and the Bradley–Terry–Luce logistic family [2, 12] are the two poles of a century-old literature; Harville [9] and Plackett [14] extend the multiplicative family to multi-entry contests, Davidson [5] adds ties, and Elo [7] and Glicko [8] are its online descendants. TrueSkill [10] shares the Thurstonian generative story with approximate Bayesian inference; we cite it for positioning only and do not test it, since its different estimation machinery would confound the link-shape question our design isolates. Mallows-type ranking models lack a latent performance scale and behave delicately in the number of alternatives [1, 6]; we stay inside the random-utility family throughout.

The lattice model — and a disclaimer. Cotton [4] computes exact win probabilities for N -entrant contests with arbitrary discretized noise, ties as dead-heat mass, and $O(N)$ cost; the construction is coherent under adding or removing entrants. That multi-entrant field coherence is the paper’s primary theorem. **This paper is not a test of Cotton’s primary theorem.** Arena data is pairwise only, and on two entrants the lattice reduces to a one-dimensional gap link (Section 3). We evaluate the link shape and tie mechanism that the lattice family induces on pairwise data—the lattice machinery is our computational vehicle, not our hypothesis. Testing field coherence itself requires listwise data that no public Arena release contains.

Arena methodology. The leaderboard fits BT over deduplicated anonymous battles with ties given half weight in each direction [3]. Our pipeline reproduces the published board to 0.18 Elo points MAE at our snapshot (Section 5). Later platform methodology adds style-controlled variants; our audit uses the raw preference log and says nothing about style adjustment.

Preference inconsistency. Positional and permutation biases in LLM judging [16], social-choice repairs [15], listwise preference objectives [11], and measured transitivity and IIA violations [18] all document departures from the axioms these models assume. Our anomaly (Section 7) is a human-vote counterpart: a violation shared by *every* static gap link we test.

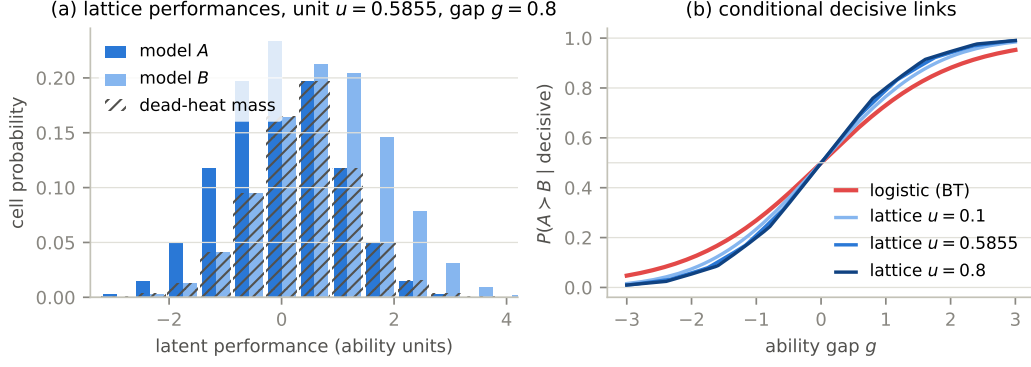


Figure 1: The lattice gap-link model. **(a)** Discretized performance distributions at gap $g = 0.8$, width $u = 0.5855$ (a coarse $L = 12$ lattice is drawn so cells are visible; fits use $L = 500$). Hatched: dead-heat mass—both performances in the same cell. **(b)** Conditional decisive links, Eq. (3). Lattice links are steeper than the logistic at the origin (slopes 0.290–0.337 vs. 0.25) and steepen as the tie band widens.

3 Two families, one theorem

3.1 Gap links

Model i carries an ability θ_i . In a battle, each side draws a performance $X = \theta + \varepsilon$ with i.i.d. noise, and the larger draw wins. Only the difference matters, so every such model is a *gap link*: with $g = \theta_A - \theta_B$,

$$\Pr(A > B) = F(g), \quad (1)$$

for an increasing F set by the noise. BT is $F = \sigma$, the logistic $\sigma(g) = (1 + e^{-g})^{-1}$ —the unique gap link satisfying IIA. Continuous noise gives ties probability zero. The two families below restore ties in opposite ways.

3.2 The lattice family: ties as dead heat, steepness as a side effect

Following Cotton [4], discretize the noise onto a lattice of cell width u . Performances land on cells $x_j = ju$; with $a_j(g) = \Pr(X_A = x_j)$ and $b_j = \Pr(X_B = x_j)$,

$$W(g) = \sum_j a_j(g) \Pr(X_B < x_j), \quad D(g) = \sum_j a_j(g) b_j, \quad L(g) = W(-g). \quad (2)$$

A tie is a dead heat: both draws in one cell (Figure 1a). The width u is the single tie parameter, fit by profile likelihood. The induced decisive link is

$$F_u(g) = \frac{W(g)}{W(g) + L(g)}. \quad (3)$$

Two facts organize everything downstream. First, F_u is steeper than the logistic at a toss-up— $F'_u(0)$ is 0.290, 0.324, 0.337 at $u \in \{0.1, 0.5855, 0.8\}$ against $\sigma'(0) = 0.25$ —and steepens as u grows. Carving dead-heat mass from the middle of the distribution sharpens what remains; the logistic's slope is unattainable at any width. Second, as a consequence, one parameter controls tie mass *and* decisive steepness. The family cannot move one without the other. We call this *entanglement* and measure its price in Section 6.4.

3.3 Davidson: ties by extension, orthogonal by theorem

Davidson [5] extends BT with a tie outcome. In gap form, with $\nu \geq 0$ and $Z_\nu(g) = 2 \cosh(g/2) + \nu$,

$$\Pr(A > B \mid g) = \frac{e^{g/2}}{Z_\nu(g)}, \quad \Pr(\text{tie} \mid g) = \frac{\nu}{Z_\nu(g)}, \quad \Pr(B > A \mid g) = \frac{e^{-g/2}}{Z_\nu(g)}. \quad (4)$$

Theorem 1 (Tie orthogonality) *For every $\nu \geq 0$, Davidson’s conditional decisive link is exactly logistic: $\Pr(A > B \mid \text{decisive}, g) = \sigma(g)$.*

Proof. Conditioning on a decisive outcome cancels Z_ν : $e^{g/2}/(e^{g/2} + e^{-g/2}) = \sigma(g)$. $\square$

Theorem 1 makes the two families a perfectly matched pair. Each has one ability per model and one tie parameter. They differ in exactly one structural respect: Davidson’s tie parameter never touches its decisive link, and the lattice’s always does. Every tie-related question in this paper comes back to this theorem—including why a decisive-only comparison of BT and Davidson would be vacuous, and what the both-bad stress test of Section 6.4 is actually measuring.

3.4 Estimation and conventions

All fits are per-vote maximum likelihood with analytic gradients; only the link differs between methods. Where an experiment replicates the production leaderboard’s outcome treatment, decisive votes count twice in their direction and each tie once in each direction through the decisive link (*half-tie* mode); where tie prediction is itself the question, the full trinomial $\{W, D, L\}$ likelihood is used. All fits of all methods anchor the same reference model (gpt-4-0613 = 0: a dated checkpoint, present with high volume in every training window; the production anchor enters too late for our earliest refits). Cross-link ability magnitudes are rescaled by $F'(0)/0.25$ so a reported unit buys the same win-probability change at a toss-up—the analogue of BT’s $400/\ln 10$ Elo constant—and are labeled convention-dependent. Sign conventions for calibration deltas are stated once per results section.

4 How much could the link matter? Effect ceilings

The question “is this link better?” has a computable prerequisite: “how much better could *any* link be?” Both links are fit to the same votes and scored on the same outcomes, so if one family were exactly true, the other’s best-fitting member would concede at most a computable amount—an amount that depends only on the two link shapes and on where the data put their ability gaps. With $\hat{P}$ the vote-weighted empirical distribution of fitted gaps and F_u a lattice truth, the logistic’s minimal concession is

$$C = \min_{s>0} \mathbb{E}_{g \sim \hat{P}} \text{KL}(\text{Bern}(F_u(g)) \parallel \text{Bern}(\sigma(g/s))). \quad (5)$$

Real matchups concentrate near toss-ups—the median fitted $|g|$ is 0.31—exactly where all increasing links agree. Evaluated across every plausible width and skew, Eq. (5) gives $C \leq 0.23 \times \text{MPD}$, where MPD is the pre-registered minimum practical difference of Section 5; the trinomial analogue for tie mechanisms is $\leq 0.31 \times \text{MPD}$ in *both* truth directions. Since per-model flexibility absorbs still more, these are upper bounds.

Three consequences. First, the honest a-priori expectation for every calibration comparison in this paper was practical equivalence—we recorded that expectation before fitting, and any verdict exceeding a ceiling would itself have demanded a misspecification investigation. Second, the ceiling is falsifiable by a second route that involves no held-out scoring at all: every gap link makes the overidentifying triple prediction $F^{-1}(p_{AC}) = F^{-1}(p_{AB}) + F^{-1}(p_{BC})$, so if the links are interchangeable on real gaps, their triple residuals must be indistinguishable too. Section 6.2 confirms this independently. Third, the methodology travels: for any two candidate models of the same data, an hour of arithmetic on the empirical design distribution can reveal—before any experiment—that the comparison cannot practically matter. We offer the ceiling as a reusable pre-analysis step, and this paper as a case study in what happens when it is taken seriously.

5 Data and pre-registered protocol

Data. The largest public Arena battle log: 1,799,991 anonymous pairwise battles, 129 models, sub-second timestamps, 2023-04-24 to 2024-08-14; the platform’s deduplication retains 1,670,250, which all fits use. Outcomes are {A wins, B wins, tie, tie (both bad)}. Quality ties are 20.45% of non-both-bad votes over the full period, up from 13.1% in the first three months—a drift that recurs throughout. The replication set is the 2025 release: 135,634 battles, 53 models, fourteen weeks (Section 6.5).

Pipeline validation. Refitting BT at the published leaderboard’s own timestamp reproduces the published ratings to MAE 0.18 Elo points (Spearman 0.99997, all 128 models). Across ten published BT-era snapshots the era-appropriate variant achieves MAE 0.18–1.01 on nine of ten; the tenth (2024-04-03) is an unexplained outlier at 2.26, reported rather than dropped.

Thresholds. Statistical significance is free at $n \approx 10^6$, so verdicts use a derived *minimum practical difference*. For decisive calibration: a uniform 10-Elo-point gap error ($\epsilon = 10 \ln 10/400$ nats) costs $\frac{1}{2}p(1-p)\epsilon^2 = 4.1 \times 10^{-4}$ at a toss-up, so $\text{MPD}_3 = 4 \times 10^{-4}$ nats/vote. For trinomial tie calibration: a one-point error in $\text{Pr}(\text{tie})$ at the observed $\sim 20\%$ level gives $\text{MPD}_4 = 3 \times 10^{-4}$. Sub-MPD effects are practically equivalent regardless of significance.

Gates and verdicts. The full pipeline passed synthetic ground-truth gates before real data: correct-sign detection at true effects down to $0.3 \times \text{MPD}$, correct positive calls at $\geq \text{MPD}$, no false positives in sub-MPD worlds, tie-parameter recovery within 10%. One gate finding matters later: under high parameter noise, a *correctly specified* steep link loses held-out to misspecified BT ($-1.41 \times \text{MPD}$)—plugin predictions from steep links are overconfident; bootstrap-calibrated standard errors (Fisher deflated by the measured 0.80; $\rho = 0.983$ against the published bootstrap) showed this reversal is inactive at our pooled noise levels. Verdicts come from a fixed classifier over thirteen rolling monthly windows (cumulative training, next-month testing): *positive* needs $|\hat{\Delta}| \geq \text{MPD}$, ten-of-thirteen sign consistency, and a window-cluster bootstrap CI (effective $N = 13$, always stated) excluding zero; *equivalence* needs the CI inside ($-\text{MPD}, +\text{MPD}$), with significant sub-MPD leans reported as leans, never wins; anything else is *inconclusive*. Post-hoc analyses are labeled and never override a pre-registered verdict. Machinery ablations (labeled) confirm the apparatus is not doing the work: pooled CIs move by under $0.004 \times \text{MPD}$ across bootstrap sizes 10^3 – 5×10^4 ; link curves are invariant to lattice resolution (to 10^{-15} in the half-width, 3×10^{-4} in the gap grid, with full-population refits identical to five decimals); and the profiled tie width moves by under 0.6% across profile-grid densities. Models absent from training are unscorable and counted: 24.8–75.8% of decisive test votes per window, a deployment fact in its own right.

6 Results

6.1 Stability: rankings do not move differently

When new models enter and the leaderboard refits, whose incumbent rankings move less? For each adjacent checkpoint pair we fit both methods on identical cumulative data and compare incumbent rankings (≥ 1000 votes; an all-models variant matches throughout).

The answer is a null with nothing hiding in it. Median paired τ_b difference +0.00026; the lattice better in 7 windows, worse in 4, tied in 2; means 0.98469 versus 0.98358; the largest single-window difference is 0.00713. The same picture holds at all three lattice widths, both horizons, both incumbent sets. No incumbent moved more than five rank positions in any window under either method, and slope-matched ability drift is 1.70–1.77 Elo-equivalent points per month for every method. Meanwhile stability itself varies from $\tau_b = 0.968$ to 0.997 across windows as the pool matures—roughly two orders of magnitude more than any between-method difference. The operative lever is sample

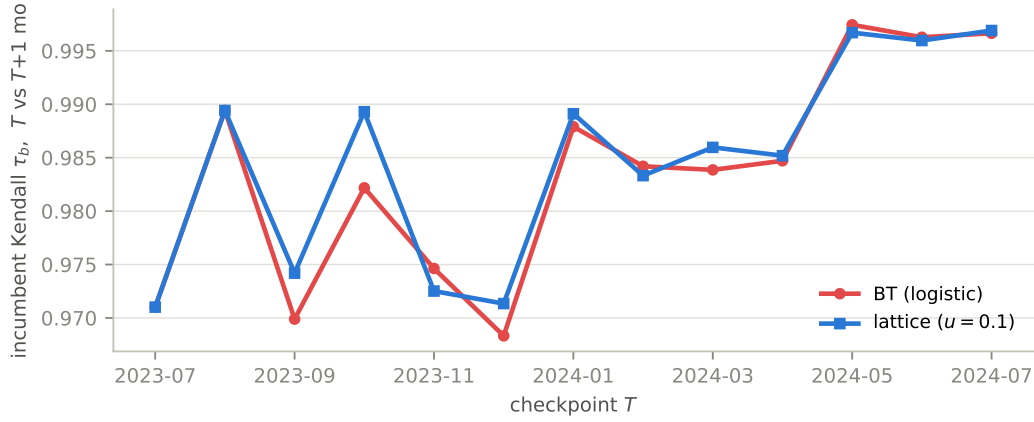


Figure 2: Month-over-month incumbent ranking stability (Kendall τ_b) for BT and the lattice link. The curves are statistically indistinguishable—median paired difference +0.00026—while both improve together as the pool matures. Source: `rq1_metrics.csv`.

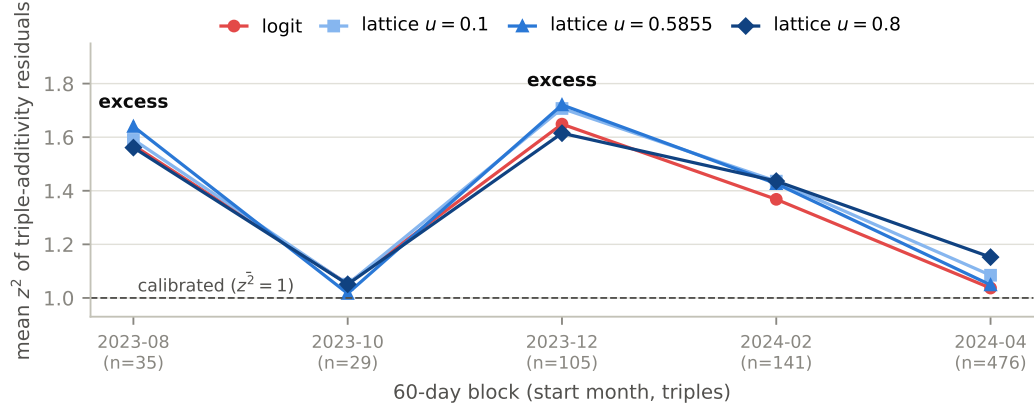


Figure 3: Mean squared triple-additivity residual per 60-day block, per link. The four links are inseparable—their spread matches the test’s synthetically measured no-power band—while two blocks show a shared excess no link explains (Section 7). Source: `rq2b_blocks.csv`.

size, not link family. A labeled post-hoc check found no modulation by entrant count or entrant vote share (all $|\rho| \leq 0.37$, $n = 13$).

Rankings do not move differently. Section 4 predicted why: where the data live, these links barely differ. That prediction can be tested directly, with no fitting at all.

6.2 Link shape: the ceiling, confirmed without fitting

Every gap link must satisfy triple additivity. For each of 786 dense stable triples (legs ≥ 100 decisive votes inside 60-day blocks, members present throughout), the standardized residual

$$z = \frac{F^{-1}(\hat{p}_{AC}) - F^{-1}(\hat{p}_{AB}) - F^{-1}(\hat{p}_{BC})}{\sqrt{v_{AB} + v_{BC} + v_{AC}}}, \quad v_{XY} = \frac{\hat{p}(1 - \hat{p})}{n_{XY} [F'(F^{-1}(\hat{p}))]^2}, \quad (6)$$

should be standard normal if the link shape is right and preferences are stable. Mean z^2 measures shape misfit scale-invariantly, so the links compete fairly.

Pooled mean z^2 : logit 1.202; lattice 1.231, 1.252, 1.279 across widths. Synthetic calibration passed (true-link $z^2 = 1.02$ – 1.03) and showed the statistic cannot separate these links at real gap

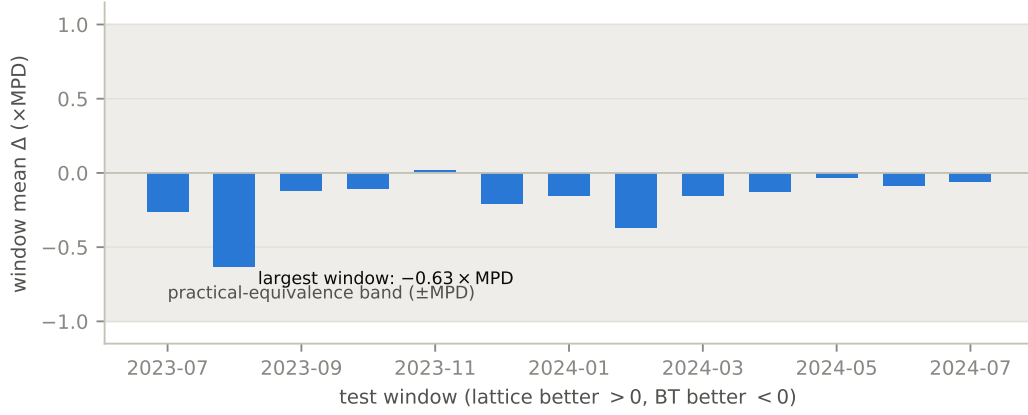


Figure 4: Per-window calibration delta (primary width $u = 0.5855$), in MPD units. Every window of thirteen lies inside the practical-equivalence band. Source: `rq3_window_table_u0.5855.csv`.

scales: in both generating directions the four links differ by 0.01–0.04, the same order as the real spread. This is the effect ceiling of Eq. (5) re-derived by an entirely different instrument—within-window ranking consistency, no held-out scoring, no plugin estimates, no time splits—and it reaches the same verdict: *at the gaps real matchups occupy, no member of this family is distinguishable from the logistic*. Two structurally different tests agreeing is stronger than either alone.

Shape is settled. But calibration on *future* votes could still differ: estimation noise interacts with steepness even when shapes agree in-sample. That is the next question.

6.3 Calibration: equivalent — and a cautionary tale

Both methods are fit per window in half-tie mode and scored by log loss on the shared scoreable decisive votes of the following month. Sign convention: $\Delta = \ell_{\text{BT}} - \ell_{\text{lattice}}$; positive favors the lattice.

Verdict: equivalence, with a sub-practical BT-leaning lean, at all three widths. Pooled $-0.121 \times \text{MPD}$, CI $(-0.191, -0.074)$ at the primary width; -0.148 $(-0.201, -0.116)$ and -0.226 $(-0.321, -0.164)$ at the others; BT better in 12–13 of 13 windows; *every* window inside the band (Figure 4; largest $0.63 \times \text{MPD}$). The lean is real and below deployment relevance—per protocol it is reported as a lean, not a win. It is largest at the steepest width but not monotone across the two shallower ones: directional, qualified evidence for the plugin-overconfidence mechanism of Section 5. The pre-registered strata agree. Recent-entrant votes: $-0.110 \times \text{MPD}$, CI $(-0.172, -0.052)$. And in the one location where the pre-analysis said a lattice-truth effect could clear MPD—the early high-noise recent-entrant stratum, predicted $+1.58 \times \text{MPD}$ under lattice-truth—the measurement is $-0.275 \times \text{MPD}$, CI $(-1.169, +0.040)$: the prediction is excluded decisively.

Two shared measurements dwarf all of this. Reliability slopes on held-out predictions are 1.46 for every method: everything is underconfident on next month’s votes by the same factor, a nonstationarity no link addresses—and uncertainty-integrated (posterior-predictive-style) prediction does not help, shifting the pooled delta only from -0.121 to $-0.117 \times \text{MPD}$ and reliability to 1.46 (labeled post-hoc; Gauss–Hermite over calibrated standard errors). Coverage is worse: up to three quarters of next-month decisive votes involve an unscorable newcomer. Drift and coverage bind; the link does not.

6.3.1 The episode that wasn’t

Our first run of this experiment told a better story, and it was wrong. It returned *inconclusive*: twelve sub-practical windows plus one at $+5.8 \times \text{MPD}$, traced to a single model that entered with two train-

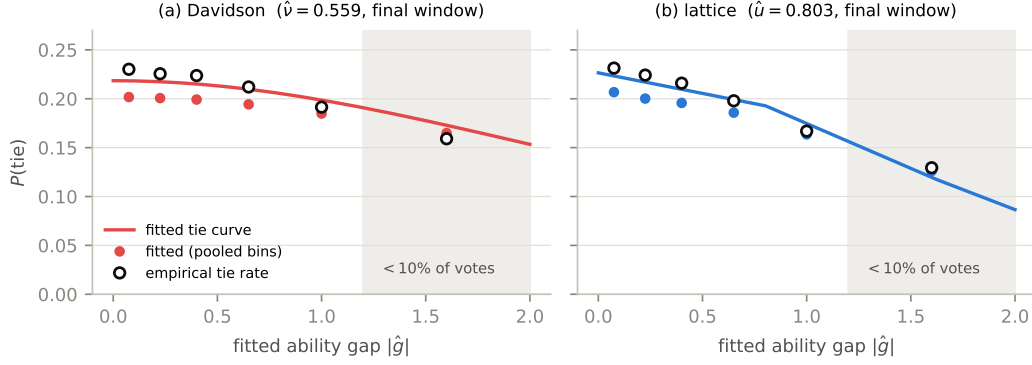


Figure 5: Fitted tie curves versus empirical tie rates, pooled with vote weights, each model binned by its own fitted gaps. The mechanisms agree at the origin, decay at structurally different rates, and the region where they separate (shaded) carries under 10% of votes. Source: `rq4_tie_curve_bins.csv`.

ing votes, which BT extrapolated to mid-table ($\hat{\theta} = +0.43$) while the steeper lattice held near zero. We initially read this as implicit shrinkage—a steep link regularizing cold starts. A reviewer-driven follow-up asked whether explicit ridge regularization could reproduce the effect, and its $\lambda \approx 0$ baseline refused to match our own stored table. The audit that followed found the cause: our earlier lattice implementation paired piecewise-linear log-likelihood values with smoothed gradients, and that inconsistency—originally noticed only as an optimizer nuisance—had effectively clamped near-zero-data models in the curve tails. With consistent interpolation the episode vanishes (old code $+5.792 \times \text{MPD}$, corrected $+0.017$, byte-identical everything else; the corrected lattice places the cold-start model at $+0.29$, barely below BT’s $+0.38$), the unfiltered result equals the old cold-start-filtered one, and the verdict above is what remains. All synthetic gates were re-certified under the corrected code.

The retraction sharpened the science. *Steepness is not implicit regularization*—but explicit regularization works exactly as the false story promised: ridge on BT shrinks the cold-start model’s estimate from $+0.38$ to 0.00 over $\lambda \in [1, 3]$ and improves that window by $\sim 5.5 \times \text{MPD}$, while established models move by at most 0.015 . Cold-start protection is real, cheap, and link-independent. And the episode is a working demonstration of the protocol: pre-registered baselines plus one skeptical follow-up caught an artifact that a lone headline run would have published.

Calibration, then, cannot separate the families. By Theorem 1, ties are the one outcome channel where they differ by construction. Is that difference measurable?

6.4 Ties: provably different shapes, indistinguishable fit — and the price of entanglement

Both models are fit per window by trinomial likelihood—abilities plus one profiled tie parameter each—and scored on next month’s {win, loss, tie} outcomes. Dead heats use the platform’s quality-tie label; “both bad” marks responses below an absolute bar, not close draws, and enters only the stress test below. Sign convention: $\Delta = \ell\ell_{\text{Davidson}} - \ell\ell_{\text{lattice}}$; positive favors the lattice; $\text{MPD}_4 = 3 \times 10^{-4}$.

Verdict: inconclusive, by one episode. Pooled $+0.838 \times \text{MPD}$, CI $(+0.017, +2.542)$; lattice better in 10/13 windows, but the estimate is carried by a single window at $+9.55 \times \text{MPD}$. Decomposition attributes that window to the tie channel on near-peer, largely same-family pairs (tie votes $+407$ nats versus -301 decisive; sibling matchups such as `gpt-4-0314` vs. `gpt-4-1106-preview` lead), where the window’s tie rate ran high and the lattice’s tie-curve shape briefly paid off. A cold-start filter does not touch it ($+0.861 \times \text{MPD}$): this episode, unlike Section 6.3.1’s, is real data behaving unusually for one month—it immediately precedes one of the two anomaly eras of Section 7, a coincidence we note without claiming.

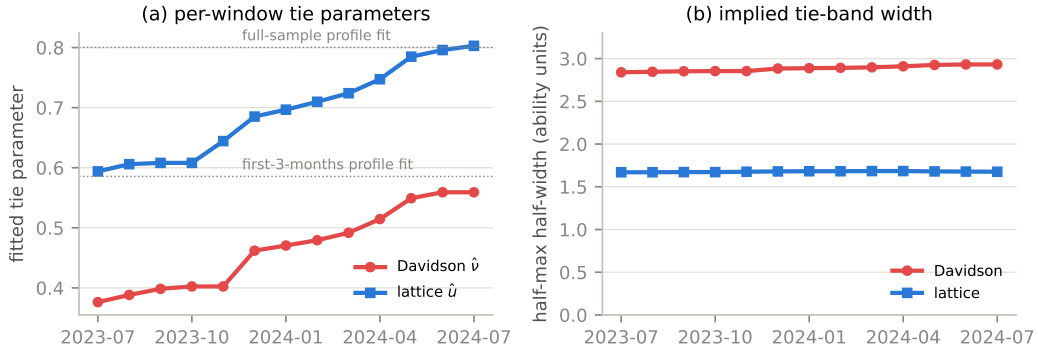


Figure 6: Drift trajectories. **(a)** Per-window profiled tie parameters rise together ($\hat{v}$: 0.376 $\rightarrow$ 0.559; $\hat{u}$: 0.594 $\rightarrow$ 0.803), tracking the tie-share drift (13.1% $\rightarrow$ 20.45%); the endpoints independently reproduce two earlier one-shot profile fits (dotted). **(b)** The implied band-width difference is stable throughout. Source: `rq4_param_trajectories.csv`.

Different shapes the data cannot discriminate. The fitted mechanisms imply nearly identical at-zero tie mass in every window (within ~ 0.01) but structurally different bands: the tie probability halves at 2.84–2.93 ability units under Davidson’s slow decay versus 1.67–1.68 under the lattice’s overlap decay—a stable $1.7\times$ difference. Held-out fit cannot pick a side (vote-weighted binned RMS 0.0225 vs 0.0237), because real matchups concentrate where the shapes agree (Figure 5). The parameters v and u carry genuinely different shape content—they are not relabelings—but this dataset cannot say which shape is right. That is a more precise statement than “equivalent,” and we intend the distinction.

Drift, cross-validated three ways. Both tie parameters rise near-monotonically across the sixteen months (Figure 6), tracking the raw tie-share drift; any single pooled tie parameter averages a moving target, which is why every experiment profiles per window. The windowed trajectory’s endpoints reproduce, to ~ 0.01 , two estimates computed independently by a different procedure months earlier—three estimation paths agreeing on one drifting quantity.

The price of entanglement, measured. Theorem 1 says Davidson’s v cannot touch its decisive link; the lattice’s u must. Forcing both models to absorb both-bad-scale tie mass ($\sim 35\%$ of votes) makes this concrete: the lattice width pins at its pre-specified profile ceiling in 11/13 windows—not for want of tie capacity but because its narrow band must over-widen to cover large-gap ties—and on identical held-out *decisive* votes its log loss degrades by $+4.75\times\text{MPD}$ (ceiling-window mean) versus $+2.50$ for the Davidson control, which isolates the shared ability-refit effect since its decisive link provably cannot move. The lattice-specific excess, $\approx 2.2\times\text{MPD}$, is the measured cost of buying tie mass with decisive accuracy. Qualifications: modest, window-heterogeneous, and the profile boundary was never stress-tested at this tie scale; the tie/both-bad category decision rests on its conceptual grounds, not on this result.

6.5 Replication: a second dataset

Every design above, transported verbatim to the 2025 Arena release (135,634 battles, 53 models, twelve weekly windows; labeled post-hoc generalization since the transport itself was not pre-registered): full-population fits agree at Spearman 0.999839 with a maximum rank move of one; the calibration comparison returns equivalence with the same sub-practical BT lean at all three widths ($-0.084/-0.175/-0.176\times\text{MPD}$; BT better in 11/12 windows); the tie comparison is again lattice-leaning inconclusive ($+0.779\times\text{MPD}$, CI $(+0.382, +1.093)$); and the profiled width barely moves over fourteen weeks (0.689 $\rightarrow$ 0.693), confirming the tie-propensity drift as a long-horizon phenomenon. Same nulls, same lean directions, same structural picture, different era and pool.

What limits next-month ranking quality here (schematic; bar lengths not to scale)

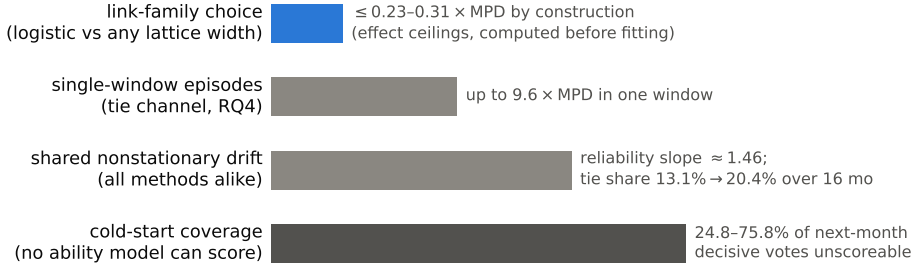


Figure 7: The error budget of this deployment, schematically. Link choice is bounded above by construction; episodes, shared drift, and cold-start coverage each exceed it by one to two orders of magnitude in their own terms. Annotated values are measured; bar lengths are not to scale.

7 The anomaly: excess dispersion no link explains

One signal in this dataset survives every attribution attempt, and we promote it deliberately: in two specific eras, *every* link we tested fails triple additivity the same way. Blocks 2023-08-22 $\rightarrow$ 10-21 and 2023-12-20 $\rightarrow$ 2024-02-18 show mean $z^2 = 1.57$ and 1.65, with 13–14% of triples beyond $|z| > 2$ against a nominal 4.6%; other blocks sit near calibration. Half-block splits localize the first episode to roughly late September–October 2023 and show the second persisting undiminished in both halves, so within-window drift is not the mechanism. Ruled out systematically: the platform’s deduplication transition (overlaps only the cleanest block); entry activity (the excess blocks have *fewer* entries and lower entrant vote share—anti-correlated); language mix, judge counts, votes per judge, tie share, duplicated-prompt share (none separates excess from clean); single-model artifacts (excess triples span models near base rates). Because the excess is invariant across link shapes, it is a property of the vote-generating process itself.

Two candidate explanations remain live. *Judge-population composition shifts*: a mixture of individually consistent judges violates additivity in aggregate. *Genuine context effects*: votes influenced by what else occupies the pool—the classical IIA-violation mechanism, appearing here as a shared failure of every IIA-like model rather than a differential one. Distinguishing them needs per-judge panel structure or listwise data that public releases do not provide. We had pre-registered an entry-proximity event study for exactly this purpose and descope it on evidence: the excess is anti-correlated with entries, so an entry-based instrument is structurally wrong for it. We state the anomaly as an open question, and as a concrete data request to preference platforms.

8 Discussion

The general principle. Every result above is one theme instantiated: richer links buy approximation accuracy, and this system’s errors are not approximation errors. Figure 7 draws the budget. The quantity a link controls is bounded at a fraction of one practical unit by Eq. (5); everything that actually limits ranking quality—shared drift miscalibration (1.46 $\times$ underconfidence for every method), cold-start coverage (up to three-quarters of votes unscorable), single-month episodes (up to $\sim 10 \times \text{MPD}$)—lives on other axes entirely. The statistical lesson travels beyond leaderboards. Reward models for RLHF are BT by construction, and listwise or richer-link refinements of the preference objective [11] face the same arithmetic: where comparison data concentrates near toss-ups and the candidate pool churns, estimation error, drift, and coverage dominate any link-shape refinement. Recommendation and matchmaking systems consuming implicit pairwise feedback, and econometric choice models fit to large observational panels, inherit the same check-list: compute the ceiling

before the comparison. Expressiveness is the right investment only when approximation error is the binding constraint, and in evolving preference systems it rarely is.

What to do instead. The audit’s constructive findings are an ordered to-do list for platform operators. First, cold starts: explicit ridge (or any shrinkage prior) on new entrants buys the protection that no link provides, at $\sim 5 \times \text{MPD}$ in the one month our data offered a natural experiment, with negligible effect on established models. Second, drift: recalibrate predictions against recent outcomes—posterior-predictive smoothing of parameter uncertainty does not substitute, because the miscalibration is shared drift, not plugin overconfidence. Third, ties: if ties matter to the product, prefer mechanisms that are orthogonal by construction (Theorem 1); a single-width dead-heat band pays for tie mass with decisive accuracy. Only after all that, consider the link.

On negative results. The pre-registration protocol was not ornamental. The ceilings set expectations before fitting; the gates certified detection power; the fixed classifier refused convenient readings twice; and the discipline caught our own numerical artifact when one skeptical baseline refused to reproduce a stored table (Section 6.3.1). We offer the protocol as the reason these nulls are informative rather than disappointing, and the artifact episode as the argument for auditing positive stories with the same energy as negative ones.

9 Limitations

This paper is not a test of Cotton [4]’s multi-entrant coherence theorem; pairwise data cannot exercise it. Conclusions concern one platform (two eras of it) and the gap-link families compared; TrueSkill-style approximate Bayesian inference was positioned but not tested. The ridge demonstration rests on one natural experiment. The dose-response evidence for plugin overconfidence is directional, not monotone. The entanglement measurement is qualified by its untested profile boundary. One validation snapshot remains an unexplained outlier. And the anomaly of Section 7 is characterized but not explained—resolving it, like testing field coherence, needs data that platforms do not currently release: judge panel structure and listwise comparisons.

10 Conclusion

We asked whether a better-motivated choice model improves a real preference leaderboard, and built the audit carefully enough to trust the answer: no—and provably, almost no such improvement was available. One theorem, one ceiling, four pre-registered comparisons, one replication, one retracted episode, and one open anomaly later, the durable lesson is the one we began with. *Richer models help when approximation error dominates estimation error.* In large, noisy, evolving preference systems—leaderboards, reward models, recommenders—estimation error, nonstationarity, and cold starts overwhelm the link function. Spend expressiveness where the errors are.

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